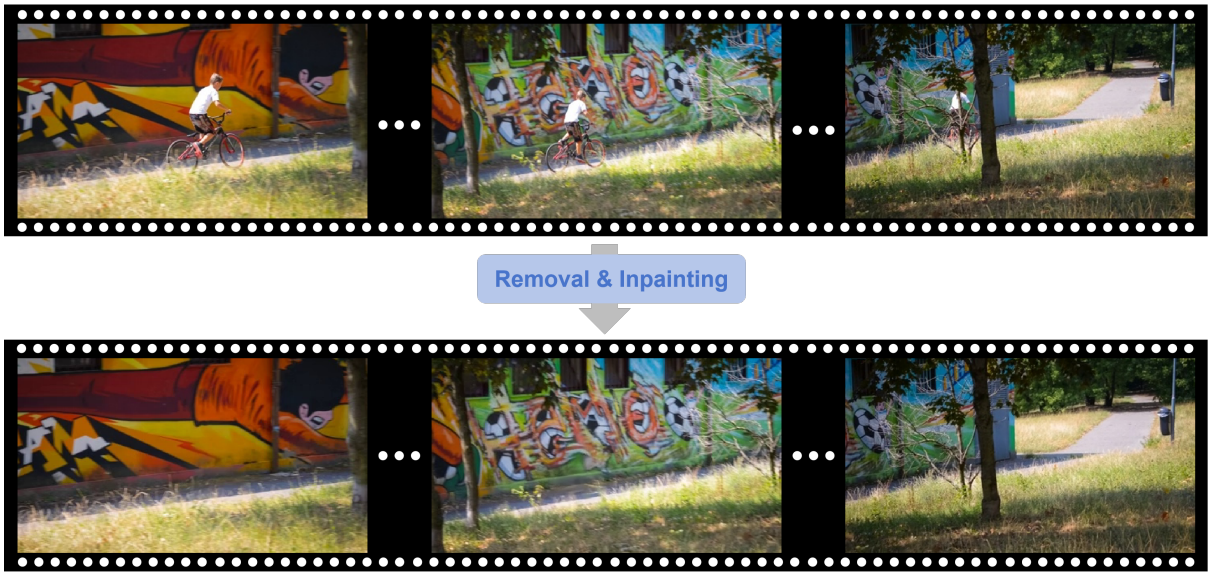
Project 3: Video Object Removal & Inpainting

AIAA 3201—Introduction to Computer Vision

Term Project Instruction

Spring 2026



1 Overview

Core Task: Given a video containing dynamic objects (e.g., pedestrians, vehicles), students are required to design a pipeline to automatically identify and remove these dynamic objects. Furthermore, the missing regions must be filled using temporal information to restore a clean background.

Key Concepts:

- **Optical Flow:** Temporal Consistency.
- **Segmentation & Object Detection:** Mask Extraction.
- **Vision Transformer:** Feature Extraction.
- **Generative AI:** Video Inpainting.

Group Size: 2 students per group.

2 Submission Requirements

- **PDF Report:**

- **Format:** Use the CVPR LaTeX Template. The length must be **6–8 pages** (excluding references).
- **ArXiv Upload (Optional):** You are required to upload your final report to **arXiv**. Remember to change the "REVIEW version" to "CAMERA-READY version" in template.
- **Content Requirements:**
  1. **Abstract & Introduction:** Background, motivation and solution overview. **Explicitly include your GitHub repository link at the end of abstract.**
  2. **Related Work:** You must review and cite **at least** all papers listed in the Reference Section (Section 4). Expanding beyond this list is encouraged.
  3. **Method:** Explain the technical roadmaps **in your own words**. Use high-quality flowcharts to assist your explanation.
  4. **Experiments:** Present quantitative results (Tables) and qualitative analysis (Figures).
  5. **Conclusion:** Summary, limitations, and future work.

- **Code (Mandatory):** Upload your code to a public **GitHub repository** with a clear `README.md` explaining how to run it. Do not submit raw code files. Include dependencies, usage, weights, and visual results in your README.
- **Demo Video (Mandatory):** You are required to submit **processed video** files for all mandatory datasets. Pack them into a zip file named `videos.zip` and upload it to Canvas.

Important Tips for Success

- **Method Flexibility:** You are encouraged to follow our suggested roadmap, but you are also free to design your own optimal method. Strict adherence to the "Implementation Roadmap" is not mandatory as long as the core tasks are fulfilled.
- **Completeness:** All three parts are **mandatory**. If you fail to complete Part 3, you must discuss the reasons and demonstrate failure cases to receive decent score. Ignoring it completely will result in an "Incomplete" grade.
- **Visuals:** Clean and aesthetic visualizations (e.g., flowcharts, rendering comparisons) are crucial for a Computer Vision course and will significantly improve your score.
- **Dataset Scope:** Completing only the mandatory datasets ensures a passing grade. To achieve a high score, you are expected to report experimental results on additional data.
- **Citations:** Ensure all related works and methods are correctly cited.

3 Guide to Start

To help you get started efficiently, we recommend the following workflow:

1. **Step 1: Literature Review.** Carefully read the papers listed in the **Recommended Reading List** (Section 4). Understanding the core concepts of Object Detection, Segmentation and Inpainting is crucial before coding.

2. **Step 2: Code Familiarization.** Clone the official repositories of the baseline methods (e.g., ProPainter). Run their provided demos to ensure your environment is set up and you understand the input/output formats.
3. **Step 3: Experimentation.** Run the methods on the **Mandatory Datasets** (see Section 6.1). Adjust code as required by the roadmap. Save your visualization results.
4. **Step 4: Writing & Submission.** Organize your results into tables and figures. Write your report using the CVPR template, citing **ALL** the references. Finally, upload your report to arXiv and submit the PDF report to Canvas.

4 Recommended Reading List

- **Object Detection:** R-CNN [4], Mask R-CNN [5], YOLO [11], DETR [2]
- **Instance Segmentation:** SAM [8], SAM2 [10], SAM3 [1], Track-Anything [15], VGGT4D [6]
- **Video Inpainting:** FGVC [3], E2FGVI [9], ProPainter [17]

5 Implementation Roadmap

5.1 Part 1: The Baseline – "Hand-crafted Approach"

Goal: Implement a complete video object removal and background restoration pipeline using basic computer vision knowledge and classic models. Understand the underlying logic from detection to restoration.

1. Mask Extraction: Instance Segmentation

- **Step 1: Semantic Extraction.** Use a classic and mature Object Detection/Segmentation model (e.g., **YOLOv8-Seg** or **Mask R-CNN** [5]) to extract masks for specific dynamic classes (e.g., Person, Bicycle).
- **Step 2: Dynamic Judgment.** For each detected object mask, calculate the motion magnitude of internal feature points using **Sparse Optical Flow** (e.g., Lucas-Kanade). Set a threshold to filter out static objects.
- *Advanced Idea:* Apply **Dilation** to smooth mask edges and ensure coverage of motion blur caused by moving objects.

2. Inpainting: Traditional Image Processing

- **Algorithm:** Use `cv2.inpaint` (based on Telea or Navier-Stokes algorithms) to fill the masked regions using spatial neighborhood pixel information.
- *Advanced Idea:* **Temporal Background Propagation**. Pure spatial repair often leads to blurring. You are encouraged to utilize the temporal redundancy of the video to prioritize "borrowing" clean background pixels from previous or subsequent frames at the same location. Use spatial inpainting only as a fallback.

Expected Result: Works well for simple videos with static backgrounds but may produce blurring in complex texture areas.

5.2 Part 2: SOTA Reproduction – "AI-driven Pipeline"

Goal: Engage with the latest open-source academic work and learn how to combine different SOTA modules to solve complex problems.

1. Dynamic Mask Extraction

Requirement: Choose **ONE** of the following methods to generate high-quality masks for dynamic objects.

- **Option A: Track Anything [15].**
  A flexible pipeline that combines Segment Anything (SAM) and XMem to segment and track objects in videos interactively.
- **Option B: SAM 2 [10].**
  Meta's unified foundation model that extends the promptable segmentation capability of SAM to the video domain with real-time performance.
- **Option C: VGGT4D [6].**
  A training-free approach that mines dynamic cues via Gram Similarity in Vision Transformer attention maps to achieve zero-shot segmentation.
- **Option D: Open Choice.**
  Any other open-source method capable of fulfilling the dynamic object segmentation requirement.

2. Inpainting: ProPainter [17]

- **Reason:** A classic work from ICCV 2023. It combines **Dual-domain Propagation** (Optical Flow + Features) and a **Sparse Transformer**. It effectively utilizes video temporal redundancy to fill large occlusions.
- **Alternatives:** Students may also choose E2FGVI [9] or FGVC [3].

Expected Result: Handles large occlusions effectively with clear background textures, outperforming Part 1.

5.3 Part 3: Exploration – "Optimization & Extension"

Goal: Identify limitations in existing methods (e.g., artifacts, inconsistent masks) and attempt improvements. Choose at least one direction below:

- **Direction A (Mask Upgrade): Introduce SAM 3 [1].**
  Idea: Since **Track Anything** relies on the original SAM, replacing its backbone with the state-of-the-art **SAM 3** [1] can significantly enhance mask quality. Alternatively, if using **VGGT4D**, use VGGT4D's outputs as prompts for SAM 3 to refine rough edges into pixel-perfect masks.
- **Direction B (Better Mask): Transplant to Stronger Foundation Models.**
  Idea: VGGT4D's limitations may stem from the underlying VGGT model. Transplant the VGGT4D [13] method to a newer 3D foundation model, such as **Pi3** [14] or **MapAnything** [7], to obtain more precise masks.
- **Direction C (Generative Inpainting): Introduce Diffusion Models.**
  Idea: ProPainter may fail when the background never appears in the video (i.e., pixels cannot be "borrowed" via propagation). Try using **Stable Diffusion [12] (Inpainting**

ControlNet [16]) to generatively repair keyframes and then propagate the results to other frames.

6 Dataset & Evaluation

6.1 Datasets

1. **Wild Video (Mandatory):** Each group must film a real-world video (e.g., a corridor with people walking) or generate a dynamic object-centric video using a model like SORA to verify generalization capability. You can download from Baidu or Google.
2. **Sample Data (Mandatory):** The provided `bmx-trees` and `tennis` scenes.
3. **DAVIS Dataset (Optional but Recommended):** A standard academic dataset providing high-quality videos and Ground Truth (GT) masks. Required for high scores.

6.2 Metrics (Mandatory)

- **Mask Quality:** IoU mean (JM) and IoU recall (JR) (Reference: VGGT4D paper). Evaluate mask accuracy for Part 1 and Part 2 (and Part 3 if improved).
- **Video Quality:** PSNR & SSIM (Reference: ProPainter paper). Calculate only for experiments with Ground Truth. **Note: You must submit the processed video files for all Mandatory Datasets (Wild Video, `bmx-trees`, and `tennis`).**
- **Qualitative Evaluation:** Visual comparison of restored frames across different methods.

References

[1] Nicolas Carion, Laura Gustafson, Yuan-Ting Hu, Shoubhik Debnath, Ronghang Hu, Didac Suris, Chaitanya Ryali, Kalyan Vasudev Alwala, Haitham Khedr, Andrew Huang, et al. Sam 3: Segment anything with concepts. *arXiv preprint arXiv:2511.16719*, 2025.

[2] Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. End-to-end object detection with transformers. In *European conference on computer vision*, pages 213–229. Springer, 2020.

[3] Chen Gao, Ayush Saraf, Jia-Bin Huang, and Johannes Kopf. Flow-edge guided video completion. In *European Conference on Computer Vision*, pages 713–729. Springer, 2020.

[4] Ross Girshick, Jeff Donahue, Trevor Darrell, and Jitendra Malik. Rich feature hierarchies for accurate object detection and semantic segmentation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 580–587, 2014.

[5] Kaiming He, Georgia Gkioxari, Piotr Dollár, and Ross Girshick. Mask r-cnn. In *Proceedings of the IEEE international conference on computer vision*, pages 2961–2969, 2017.

[6] Yu Hu, Chong Cheng, Sicheng Yu, Xiaoyang Guo, and Hao Wang. Vggt4d: Mining motion cues in visual geometry transformers for 4d scene reconstruction. *arXiv preprint arXiv:2511.19971*, 2025.

[7] Nikhil Keetha, Norman Muller, Johannes Schonberger, Lorenzo Porzi, Yuchen Zhang, Tobias Fischer, Arno Knapitsch, Duncan Zauss, Ethan Weber, Nelson Antunes, et al. Mapanything: Universal feed-forward metric 3d reconstruction. *arXiv preprint arXiv:2509.13414*, 2025.

[8] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. Segment anything. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4015–4026, 2023.

[9] Zhen Li, Cheng-Ze Lu, Jianhua Qin, Chun-Le Guo, and Ming-Ming Cheng. Towards an end-to-end framework for flow-guided video inpainting. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 17562–17571, 2022.

[10] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, et al. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024.

[11] Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. You only look once: Unified, real-time object detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 779–788, 2016.

[12] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10684–10695, 2022.

[13] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. Vggt: Visual geometry grounded transformer. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 5294–5306, 2025.

[14] Yifan Wang, Jianjun Zhou, Haoyi Zhu, Wenzheng Chang, Yang Zhou, Zizun Li, Junyi Chen, Jiangmiao Pang, Chunhua Shen, and Tong He. Pi3: Permutation-equivariant visual geometry learning. *arXiv preprint arXiv:2507.13347*, 2025.

[15] Jinyu Yang, Mingqi Gao, Zhe Li, Shang Gao, Fangjing Wang, and Feng Zheng. Track anything: Segment anything meets videos. *arXiv preprint arXiv:2304.11968*, 2023.

[16] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 3836–3847, 2023.

[17] Shangchen Zhou, Chongyi Li, Kelvin CK Chan, and Chen Change Loy. Propainter: Improving propagation and transformer for video inpainting. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 10477–10486, 2023.